

Module 06: Classless Addressing (CIDR) and Supernetting

Classless Inter-Domain Routing (CIDR), 3 Mandatory Block Allocation Rules, Supernetting Mechanics, and Longest Prefix Match (LPM) Routing Forwarding Algorithm.

CIDR Slash Notation, Block Rules & Supernetting (Route Aggregation)

1. CIDR Address Breakdown: 200.16.0.0 / 22

Network Prefix: n = 22 Bits (Mask: 255.255.252.0)

Host Bits: 32 - n = 10 Bits (1024 IPs)

3 Mandatory Rules for CIDR Allocation

- Rule 1 (Contiguous): All addresses must be continuous.
- Rule 2 (Power of 2): Total IPs N = 2^(32 - n) (e.g. 1024, 2048).
- Rule 3 (Divisible): First IP (in decimal) must be divisible by N.

Supernetting: 4 Class C Networks Aggregated

198.16.0.0/24

198.16.1.0/24

198.16.2.0/24

198.16.3.0/24

↓ Merged into Single Routing Table Entry ↓

Supernet: 198.16.0.0 / 22 (Mask: 255.255.252.0)

Longest Prefix Match (LPM) Routing Table Decision Rule

Destination Network	Prefix Length	Next Hop / Interface	Packet: 200.24.16.99 (Evaluation)
200.24.0.0	/16 (255.255.0.0)	Interface eth0	Matches (16-bit match)
200.24.16.0	/20 (255.255.240.0)	Interface eth1	Matches (20-bit match)
200.24.16.64	/26 (255.255.255.192)	Interface eth2	Selected! Longest Prefix (26 bits)

1. 3 Mandatory Rules of CIDR Allocation

Rule 1: Addresses in a block must be contiguous.

Rule 2: Total count of addresses must be power of 2 (N = 2^(32 - n)).

Rule 3: First address (in decimal integer) must be divisible by block size N.

2. Subnetting vs Supernetting Comparison

Parameter	Subnetting	Supernetting
Objective	Divide large network into smaller subnets	Combine smaller contiguous networks
Prefix Mask	Prefix increases (/24 to /26)	Prefix decreases (/24 to /22)
Routing Table	Internal routing precision	Compacts global Internet routing table

3. Longest Prefix Match (LPM)

When multiple routing prefixes match an incoming destination IP address, the router selects the entry with the **highest prefix length (most specific mask)**.